

The DevOps Engineer's Docker Interview Handbook-Part1

1. Why was Docker created?

- Docker was created to solve the 'works on my machine' problem by packaging applications, runtime, libraries, dependencies, and configuration into portable images that run consistently across development, testing, staging, and production environments.

2. What is Docker?

- Docker is an open-source containerization platform that builds, ships, and runs applications inside lightweight, isolated containers.

3. What is containerization?

- Containerization packages an application and all required dependencies into a single unit that can run consistently on any system with Docker installed.

4. Docker Image vs Docker Container

- A Docker Image is a read-only template containing the application and dependencies. A Docker Container is a running instance of that image with its own writable layer.

5. Docker vs Virtual Machine

- Containers share the host kernel, making them lightweight and fast. Virtual machines include a complete guest operating system, making them heavier and slower to start.

6. Explain Docker Architecture.

- The Docker Client sends commands to the Docker Daemon, which builds images, creates containers, manages networks and volumes, and communicates with Docker registries.

7. What is Docker Daemon?

- The Docker Daemon is the background service responsible for building images, running containers, managing networks, volumes, and other Docker resources.

8. What happens when docker run is executed?

- Docker checks for the image locally, pulls it if necessary, creates a writable container layer, starts the container, and executes the container's main process.

9. Why are Docker Images read-only?

- Read-only images ensure consistency, reusability, and allow multiple containers to share the same image safely without modifying it.

10. What is Copy-on-Write?

- When a container modifies a file from the image, Docker copies only that file into the container's writable layer while preserving the original image.

11. What are Docker image layers?

- Each Dockerfile instruction creates a new image layer. Docker reuses unchanged layers to improve build performance and storage efficiency.

12. What is the writable layer?

- It is the top layer added to every container where runtime changes such as logs and temporary files are stored.

13. Why do containers start faster than VMs?

- Containers do not boot a complete operating system; they share the host kernel, reducing startup time and resource usage.

14. What is a Dockerfile?

- A Dockerfile is a text file containing instructions used to build a Docker image automatically.

15. Purpose of FROM instruction?

- The FROM instruction specifies the base image from which a new Docker image is built.

16. Difference between RUN and CMD?

- RUN executes during image build, while CMD specifies the default command executed when the container starts.

17. What is ENTRYPOINT?

- ENTRYPOINT defines the fixed executable that always runs when the container starts. CMD commonly provides default arguments to it.

18. Why use WORKDIR?

- WORKDIR sets the default working directory for subsequent Dockerfile instructions, improving readability and maintainability.

19. Why use COPY?

- COPY transfers files from the local build context into the Docker image.

20. Why use Docker Compose?

- Docker Compose manages multi-container applications using a single YAML file, simplifying deployment, networking, and volume management.

21. What is depends_on?

- depends_on controls container startup order but does not guarantee that the dependent service is fully ready.

22. What is a Docker Volume?

- A Docker Volume provides persistent storage that remains available even after containers are removed.

23. What is Docker Registry?

- A Docker Registry is a centralized repository used to store and distribute Docker images.

24. Docker Hub vs Docker Registry

- Docker Hub is Docker's official public registry, whereas Docker Registry is the general term for any service that stores Docker images.

25. Why should latest not be used in production?

- The latest tag is only a label and may point to different image versions over time. Production environments should use explicit version tags for predictable deployments.